

Question 1 – Household Budget (Pie Chart)

```
# Household Budget
```

```
category <- c("Housing", "Transportation", "Food", "Entertainment")
```

```
percent <- c(40, 20, 30, 10)
```

```
# Pie Chart
```

```
pie(percent,
```

```
  labels = paste(category, percent, "%"),
```

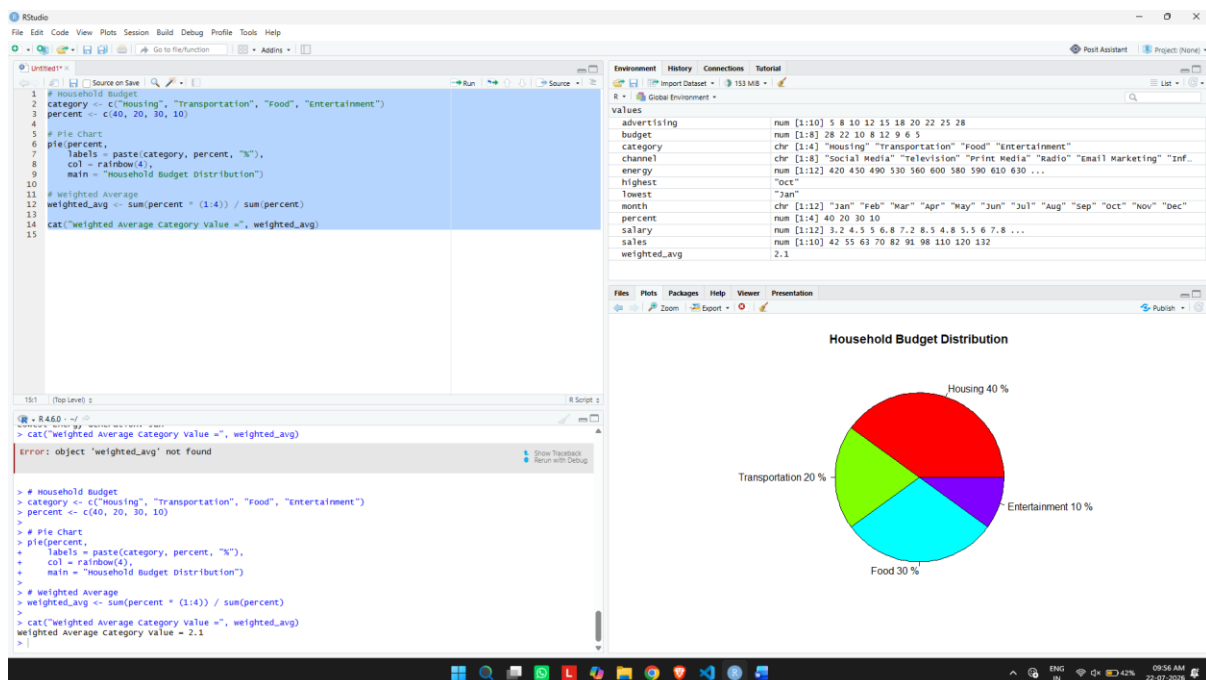
```
  col = rainbow(4),
```

```
  main = "Household Budget Distribution")
```

```
# Weighted Average
```

```
weighted_avg <- sum(percent * (1:4)) / sum(percent)
```

```
cat("Weighted Average Category Value =", weighted_avg)
```



Question 2 – Student Grades (Bar Chart)

Grade Distribution

```
grades <- c("A", "B", "C", "D", "E")
```

```
students <- c(6, 8, 9, 5, 2)
```

Bar Chart

```
barplot(students,
```

```
  names.arg = grades,
```

```
  col = "lightblue",
```

```
  main = "Student Grade Distribution",
```

```
  xlab = "Grades",
```

```
  ylab = "Number of Students")
```

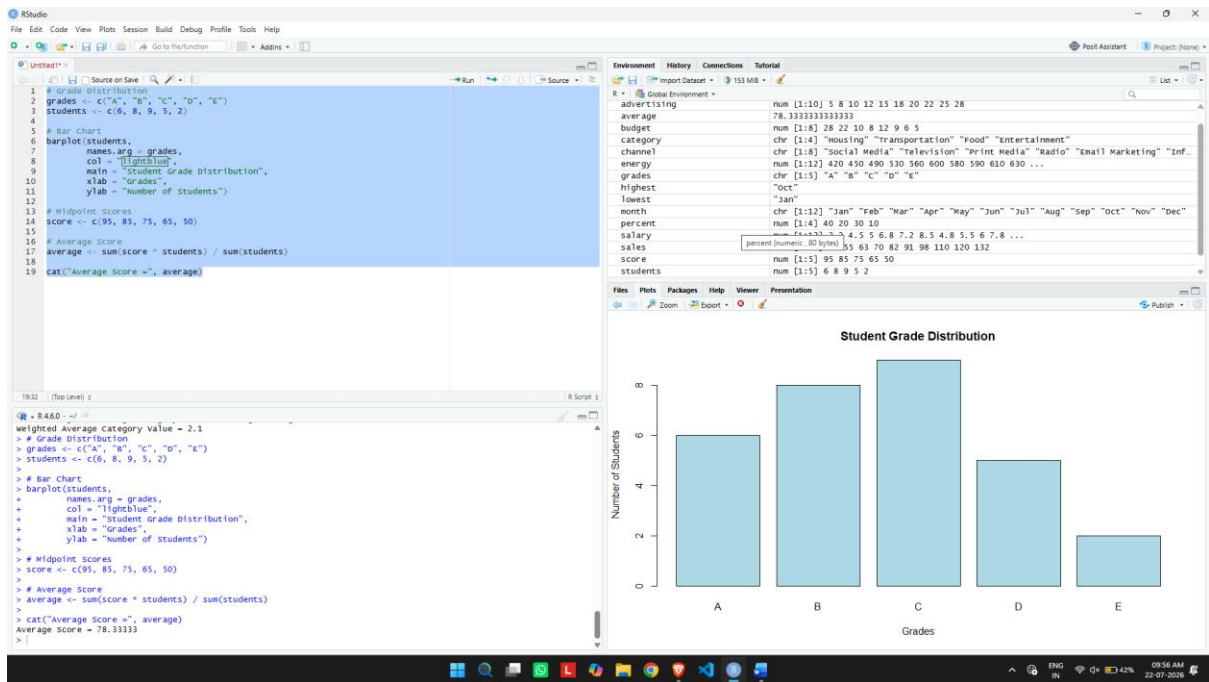
Midpoint Scores

```
score <- c(95, 85, 75, 65, 50)
```

Average Score

```
average <- sum(score * students) / sum(students)
```

```
cat("Average Score =", average)
```



Question 3 – Favourite Fruit (Bar Chart)

Favourite Fruits

```
fruit <- c("Apple", "Banana", "Mango", "Orange", "Grapes")
```

```
votes <- c(40, 35, 60, 25, 20)
```

Bar Chart

```
barplot(votes,
```

```
      names.arg = fruit,
```

```
      col = "orange",
```

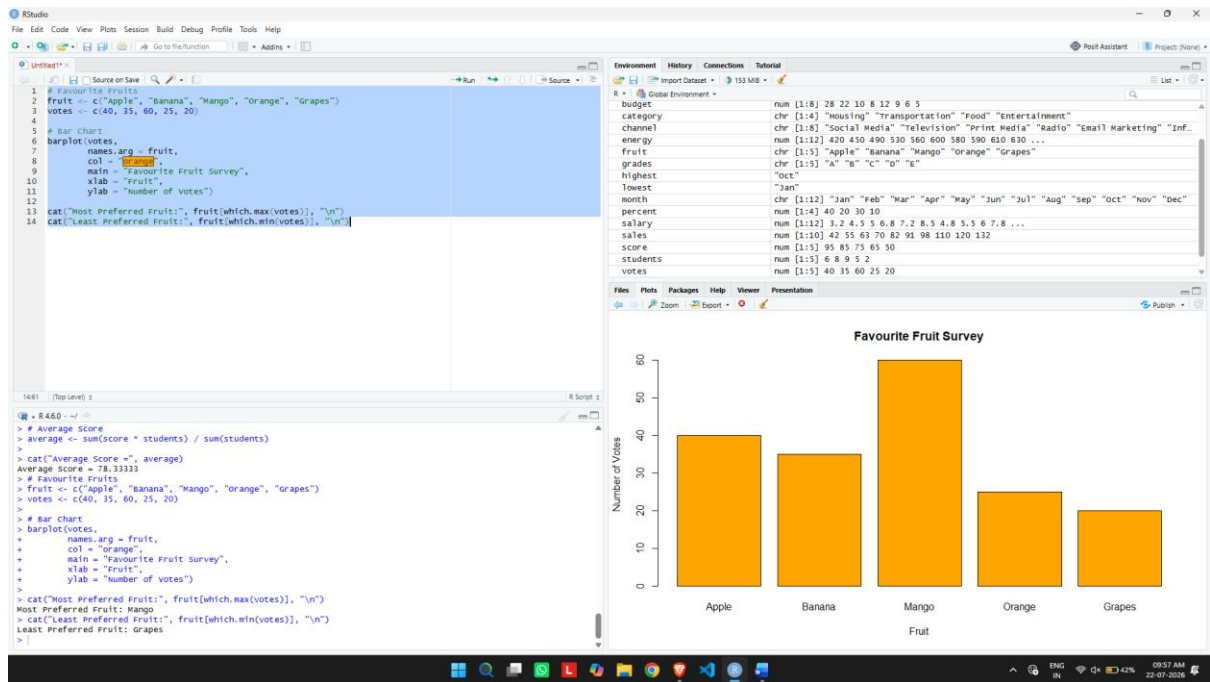
```
      main = "Favourite Fruit Survey",
```

```
      xlab = "Fruit",
```

```
      ylab = "Number of Votes")
```

```
cat("Most Preferred Fruit:", fruit[which.max(votes)], "\n")
```

```
cat("Least Preferred Fruit:", fruit[which.min(votes)], "\n")
```



Question 4 – Employees by Department (Pie Chart)

Department Distribution

```
department <- c("Marketing", "HR", "Finance", "IT", "Operations")
```

```
percent <- c(25, 15, 20, 25, 15)
```

Pie Chart

```
pie(percent,
```

```
  labels = paste(department, percent, "%"),
```

```
  col = rainbow(5),
```

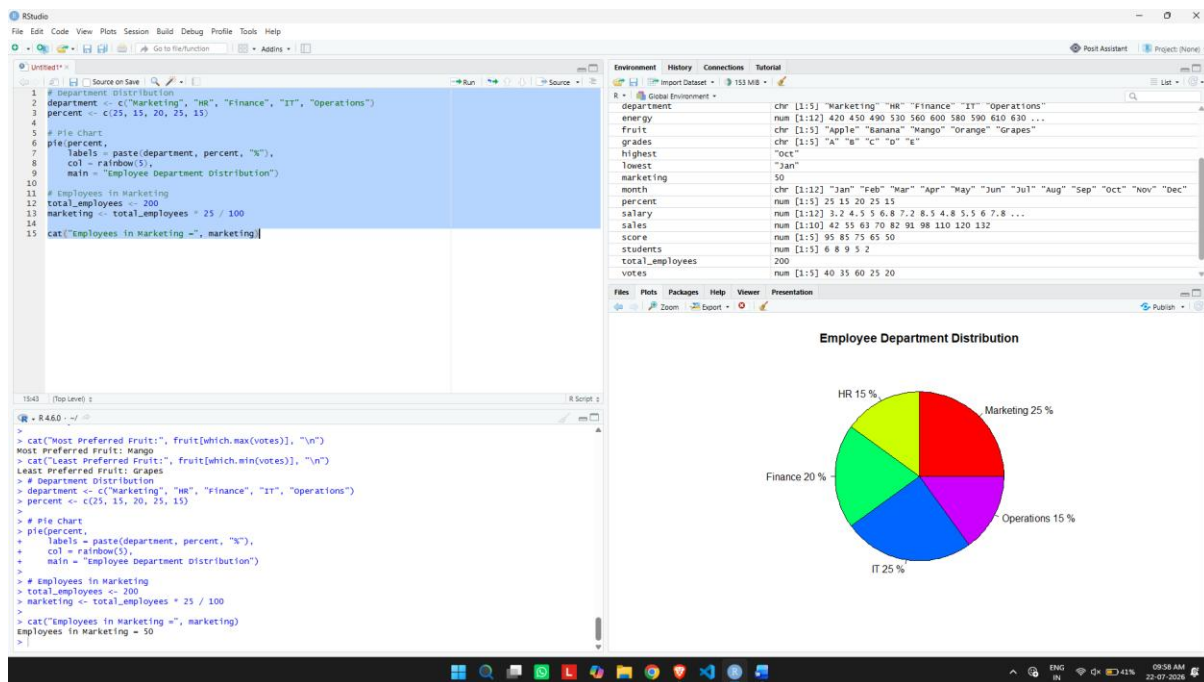
```
  main = "Employee Department Distribution")
```

Employees in Marketing

```
total_employees <- 200
```

```
marketing <- total_employees * 25 / 100
```

```
cat("Employees in Marketing =", marketing)
```



Question 5 – Weekend Activities (Pie Chart)

Weekend Activities

activity <- c("Video Games",

"Reading",

"Cooking",

"Painting",

"Music")

percent <- c(20, 35, 20, 10, 25)

Pie Chart

pie(percent,

labels = paste(activity, percent, "%"),

col = rainbow(5),

main = "Weekend Activities of Employees")

```
cat("Most Common Activity:",
    activity[which.max(percent)])
```

